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Transfer of Durable Stripe Rust Resistance Gene *Yr39* into Four Chinese Elite Wheat Cultivars Using Marker-Assisted Selection

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Abstract: Wheat gene *Yr39* confers durable high-temperature adult-plant (HTAP) resistance to stripe rust caused by *Puccinia striiformis* f. sp. *tritici*, one of the most destructive diseases of wheat worldwide. The objective of this study was to transfer *Yr39* into four Chinese elite wheat cultivars. Backcross inbred line populations were developed from four Chinese wheat cultivars Chuanmai 42 (CM42), Bainong Aikang 58 (AK58), Han 6172 (H6172) and Zhengmai 9023 (ZM9023) crossed with a *Yr39* single-gene line. The F_1, BC_1F_1 and BC_1F_6 lines were genotyped using resistance gene analogs polymorphism (RGAP) markers *Xwgp36*, *Xwgp44* and *Xwgp43*, which are closely linked to *Yr39*. Progeny lines selected with the markers for *Yr39* were evaluated in the field for stripe rust resistance and agronomic traits including plant height, tiller numbers, spike grain numbers and thousand-grain weight. Eleven lines were selected with stripe rust resistance and desirable agronomic traits. These lines with production potential can be used for further testing in various wheat production regions and as germplasm resources for breeding new wheat cultivars with durable stripe rust resistance, high yield, and adaptation to different production environments.

Keywords: WHEAT; stripe rust; high-temperature adult-plant; *Yr39*; disease resistance breeding; MAS

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1. Introduction

Wheat stripe rust, caused by the fungal pathogen *Puccinia striiformis* f. sp. *tritici* (*Pst*), is an important disease worldwide. In China, the disease occurs every year, and severe epidemics causing huge economic losses are frequent. The average annual disease area is estimated at about 267,000 hectares with countrywide yield losses of 0.5% to 5% in normal years, and 5% to 25% in epidemic years [1,2]. In recent years, with the prevalence of *Pst* race CYR34, elite wheat cultivars such as CM42, AK58, H6172, and ZM9023 have gradually lost stripe rust resistance [3–5]. Breeding stripe rust resistant cultivars is one of the most economical and effective ways to control the disease [6,7]. It is urgent to develop new wheat cultivars with resistance to stripe rust and excellent agronomic traits using existing germplasm resources [2].

Developing disease-resistant and high-yielding wheat cultivars is the major task for every breeding program [8]. This is a long process if using only the traditional phenotypical evaluation and selection. Marker-assisted selection (MAS) can be used to speed up the

process. Molecular markers specifically or closely linked to a target gene are useful in MAS [9]. Several molecular marker techniques have been developed and used in MAS, as well as mapping genes [10–13]. Since 2000, resistance gene analogs polymorphism (RGAP) and simple sequence repeats (SSR) markers have gradually become widely used [14]. RGAP primers are designed based on the conservative motifs of disease resistance genes, and such markers may be related to the target gene. However, due to the conserved sequences of the primers, a single pair of RGAP primers may amplify many fragments, which reduces the specificity [15]. Because SSR markers are mostly codominant, highly polymorphic, and easy to use, this technique has been widely used in genetic research and breeding [14]. Yin et al. developed the RGAP marker *Xrga-1* closely linked to wheat stripe rust resistance gene *YrZH84* and verified the reliability of the marker by testing 58 Chinese wheat cultivars [16]. Using SSR markers *Xcfd73*, *Xwgm120*, *Xbarc87* and *Xbarc133* closely linked to stripe rust resistance QTL *QYr.nafu-2BL* and *QYr.nafu-3BS*, Hu et al. successfully introduced two QTL from spring wheat cultivar P9897 into three main Chinese cultivars and selected 13 resistant lines with excellent agronomic traits [17]. Zhang et al. evaluated eight SSR markers linked to *Yr69* for their effectiveness in wheat breeding and found that some of the markers were more useful than the others [18]. These studies further demonstrated the feasibility of using RGAP and SSR markers in MAS for developing stripe rust resistant wheat cultivars.

Lin and Chen mapped HTAP resistance gene *Yr39* to chromosome 7BL in spring wheat cultivar Alpowa using 14 RGAP and 2 SSR markers [19]. The length of *Yr39* genetic linkage is 59.8 cm, and its flanking markers are *Xwgp36* and *Xwgp45*. Coram et al. used the transcriptome approach to determine the molecular mechanism of the non-race-specific HTAP resistance controlled by *Yr39* in comparison with race-specific resistance [20]. The single *Yr* gene line AvS/Alpowa F_7-71 for *Yr39* was derived from a cross Avocet Susceptible (AvS)/Alpowa. Through testing in the greenhouse and field, Zhou et al. found that this *Yr39* line was consistently resistant at the adult-plant stage when inoculated with Chinese predominant *Pst* races, including CYR31, CYR32 and CYR33 [21]. At present, the effort for using HTAP resistance in China is limited, and no Chinese wheat cultivars have been reported to carry *Yr39*.

The objective of the present study was to transfer into four elite Chinese wheat cultivars using marker-assisted selection (MAS) to obtain resistance cultivars with excellent agronomic traits. Progeny lines selected by molecular markers for *Yr39* were confirmed in the field for stripe rust resistance and further selected for desirable agronomic traits. The selected lines with HTAP resistance in the elite Chinese wheat backgrounds can be used for further tests in various wheat growing regions for potential use in production and also can be used in breeding programs as new resources of durable stripe rust resistance.

2. Materials and Methods

2.1. Plant Materials

The spring wheat line AvS/AlpF_7-71 carrying *Yr39* (hereinafter referred to as the *Yr39* line) was used as the donor of *Yr39* [20,21]. Elite Chinese wheat cultivars CM42 (National-approved wheat NO. 2004002), AK58 (NO. 2005008), H6172 (NO. 2003036), and ZM9023 (NO. 2003027) released in 2004, 2005, 2003 and 2003, respectively, were used as the female and recurrent parents in crossing with the *Yr39* line, and as references in evaluation for stripe rust resistance and agronomic traits. With excellent agronomic traits, these cultivars were resistant to stripe rust before and when released. With high yield, disease resistance, and good quality, CM42 has been widely grown in the winter wheat areas in the upper reaches of the Yangtze River. AK58 is a facultative winter wheat with strong tillering ability, compact plant type and strong lodging resistance, and has been widely grown in the Huanghuai winter wheat area. H6172 is a facultative winter wheat developed by the Handan Academy of Agricultural Sciences in Hebei Province, and it has a high number of spikes, bright white grain and resistance to stem rust. ZM9023 has the characteristics of quick grain filling and early maturity, and it has been widely grown in several provinces (http://202.127.42.47:6010/SDSite/Home/Index (accessed on 17 February 2022)). CM42

has been reported to have *Yr26* for race-specific all-stage resistance (ASR) [22], and ZM9023 may have *Yr29* and *Yr30* for adult-plant resistance (APR) to stripe rust [23]. The stripe rust resistance genes in AK58 and H6172 are unknown. The stripe rust resistances of the four cultivars are no longer effective [8,24].

2.2. Development of Breeding Population

The *Yr39* line was used as the paternal parent, and CM42, AK58, H6172, and ZM9023 were used as maternal parents in the initial crosses to develop breeding populations for transferring the gene. The crosses and backcrosses were made in the greenhouse and experimental field (N 31°69′09.96″, E 104°66′99.13″) of the Wheat Research Institute of Southwest University of Science and Technology. In 2016, F_1 plants from crosses CM42/*Yr39*, AK58/*Yr39*, H6172/*Yr39* and ZM9023/*Yr39*, and BC_1F_1 plants from backcrosses CM42//CM42/*Yr39*, AK58//AK58/*Yr39*, H6172//H6172/*Yr39* and ZM9023//ZM9023/*Yr39* were obtained in the greenhouse and tested with the linkage markers *Xwgp36* and *Xwgp44* of *Yr39*. In the 2016–2021 growing seasons, the progeny lines were planted in the field for advancing the generations and evaluated for stripe rust resistance and agronomic traits. Mingxian 169 (MX169) was used as a susceptible control in the stripe rust tests.

2.3. Field Evaluation

Sichuan province is one of the regions with high numbers of *Pst* races in China [25]. Mianyang is an important wheat stripe rust epidemic area in the province, where *Pst* can overwinter. In the last few years, the predominant popular races were CYR32, CYR33, and CYR34 (Zhou Xinli unpublished data).

In the 2016–2017 and 2017–2018 growing seasons, BC_1F_2 and BC_1F_3 seeds of the four crosses were planted, respectively, in rows of 200 cm in length with 80–100 seeds in each row and 30 rows for each cross. The seeds of each generation were bulk-harvested for each cross. In the 2018–2019 growing season, the BC_1F_4 populations were evaluated for stripe rust resistance and agronomic traits in the experimental field. Lines with undesirable agronomic traits [plant height (PH) $\geq$ 100 cm, number of tillers (NT) < 3, spike grain number (SGN) < 40] using the parents as the references and/or susceptible to stripe rust (IT 7–9) were discarded. In the 2019–2021 growing seasons, the selected BC_1F_5 lines of each cross were evaluated for stripe rust resistance and agronomic traits using a randomized complete-block design experiment with three replications. In each replicate, each line was planted in three rows with 40 seeds in each row of 100 cm long and 25 cm between rows, and a 40 cm aisle between plots. MX169 was planted in the borders of the field as spreaders.

Stripe rust on each row was evaluated from heading (Z50) to grain filling (Z80) [26]. Infection type (IT) was recorded following the 0–9 scale [27], with 0 considered immune, 1 to 3 resistant, and 4 to 6 moderately resistant, and 7 to 9 susceptible. Disease severity was recorded as 0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95 or 100% for the percentage of leaf area infected by *Pst*. The agronomic traits were evaluated during the grain-filling period. For each line, 30 individual plants were randomly selected to measure PH, NT, SGN, and thousand-grain weight (TGW). All screened lines were named with the abbreviation of Southwest University of Science and Technology, SWUST.

2.4. DNA Extraction and MAS

DNA was extracted from leaf samples collected at the three-leaf stage. Samples in a 2 mL centrifuge tube with steel balls were put immediately into liquid nitrogen for freezing and ground using a Geno/Grinder 2010 tissue grinder (SPEX SamplePrep, Metuchen, NJ, USA). The Cetyltrimethylammonium bromide (CTAB) method was used to extract genomic DNA from the fresh leaf samples [28]. A NanoDrop ND-1000 spectrophotometer (Thermo Scientific, Wilmington, NC, USA) was used to determine the quantity and quality of the stock DNA solution. Deionized distilled water was added to adjust the DNA concentration to 50 ng/μL, and the solution was transferred to a 96-well plate for storage.

The parents, F_1, BC_1F_1, and BC_1F_5 plants were tested with two RGAP markers, *Xwgp36* and *Xwgp44* [19]. The selected BC_1F_6 lines and five parents were tested using three RGAP markers, *Xwgp36*, *Xwgp43* and *Xwgp44* (Table 1). *Xwgp45* did not amplify the specific band in the parents, so *Xwgp45* was replaced with *Xwgp44* which was 1.9 cm away from *Xwgp45*. The primers used in this study were synthesized by Shenggong Biological Engineering (Shanghai) Co., Ltd., The PCR reaction was in 15 µL volume, containing 1.2 µL 50 ng/µL DNA template 5 µL 10× PCR buffer, 1.2 µL 2.5 mM dNTPs, 1.2 µL forward primer, 1.2 µL reverse primer, 0.12 µL 5 units/µL Taq enzyme, and 5.08 µL deionized water. The PCR amplification procedure included pre-denature at 94 °C for 5 min; followed by denaturation at 94 °C for 1 min, annealing at 45 °C for 1 min, 72 °C extension for 2 min, 40 cycles; and 72 °C extension for 7 min, and then kept at 16 °C for later use. The PCR amplification products were separated using 6% polyacrylamide gel electrophoresis (constant pressure 1200 V, 2 h), and stained with 0.1% silver nitrate solution for 20 min and visualized for the expected bands. The *Yr39* line was included in these experiments as the positive control.

Table 1. Three RGAP primer sequences closely linked to wheat resistance to stripe rust gene *Yr39*.

Marker	Primer	Sequence (5′–3′)	Product Size (bp)	Reference
Xwgp36	*Pto kin1*	GCATTGGAACAAGGTGAA	830	Lin and Chen [19]
	RLK For	GAYGTNAARCCIGARAA		
Xwgp43	*Pto kin1*	GCATTGGAACAAGGTGAA	820	Lin and Chen [19]
	PtoFen-S	ATGGGAAGCAAGTATTCAAGGC		
Xwgp44	*S2-INV*	CAICAIAAIGGITGIGGIGG	875	Lin and Chen [19]
	NLRR-INV2	TCAGGCCGTGAAAAATAT		

2.5. Statistical Analysis

The data of stripe rust and agronomic traits were analyzed using software Origin 2021 (https://www.originlab.com/index.aspx?go=Products/Origin (accessed on 22 February 2022)). Variance analysis was performed using the "AVO" tool in IciMapping V4.2 (http://www.isbireeding.net/ (accessed on 22 February 2022)).

3. Results

3.1. Development of Backcross Inbred Line Populations

To develop the *Yr39* transfer populations, 22 F_1 plants were selected from 51 plants tested based on the presence of both *Xwgp36* and *Xwgp44* markers, including 5 from CM42/*Yr39*, 4 from AK58/*Yr39*, 7 from H6172/*Yr39*, and 6 from ZM9023/*Yr39*. The selected F_1 plants were backcrossed with their respected recurrent parents to obtain the BC_1F_1 population. After testing with *Xwgp36* and *Xwgp44*, 35 BC_1F_1 plants were identified having the *Yr39* markers, including 8 from CM42//CM42/*Yr39*, 5 from AK58//AK58/*Yr39*, 16 from H6172// H6172/*Yr39*, and 6 from ZM9023//ZM9023/*Yr39*. The BC_1F_1 plants were selfed to produce BC_1F_2 seeds in the greenhouse and advanced to the BC_1F_4 generation in the field. In the preliminary screening of BC_1F_4 lines in the field, 71 lines with stripe rust resistance and desirable agronomic traits were finally selected, including 17 from CM42//CM42/*Yr39*, 11 from AK58//AK58/*Yr39*, 26 from H6172//H6172/*Yr39*, and 17 from ZM9023//ZM9023/*Yr39* (Table 2).

Table 2. Four cultivar (CM42, AK58, H6172 and ZM9023) crosses with *Yr39* showed positive lines in different generations as detected by *Yr39*-linked RGAP markers.

	CM42/*Yr39*		AK58/*Yr39*		H6172/*Yr39*		ZM9023/*Yr39*		Total	
Generations	Tested	Positive	Tested	Positive	Tested	Positive	Tested	Positive	Tested	Positive
F_1	12	5	10	4	19	7	10	6	51	22
BC_1F_1	17	8	13	5	22	16	15	6	67	35
BC_1F_5	17	8	11	6	26	14	17	4	71	32
BC_1F_6	17	8	11	5	26	5	17	2	71	20

3.2. Evaluation of the Selected Lines for Resistance to Stripe Rust

In the field tests, the *Yr39* line had IT 2 and DS 10%, CM42 had IT 7 and DS 60%, AK58 had IT 8 and DS 70%, H6172 had IT 6 and DS 70%, and ZM9023 had IT 7 and DS 60% (Figure 1). When the two-year field data were combined, the 17 lines from CM42//CM42/*Yr39* and the 11 lines from AK58//AK58/*Yr39* were highly resistant to stripe rust (IT 1–3). Of the 26 lines from H6172//H6172/*Yr39*, 17 were highly resistant with IT 1–3 and 9 moderately resistant with IT 4–6. Of the 17 lines from ZM9023//ZM9023/*Yr39*, 15 were highly resistant and 2 moderately resistant (Table 3). Due to the selection of resistance phenotypes in BC_1F_4 generation, there were no susceptible lines from all crosses, and the DS values of these lines were less than 50% except for SWUST-43 (IT 6, DS 60%) from H6172//H6172/*Yr39*.

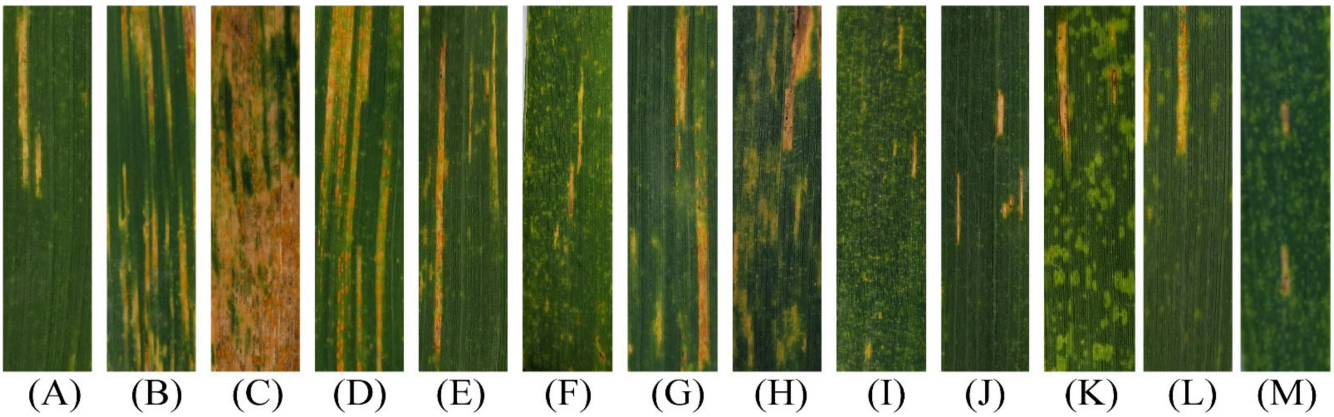

Figure 1. Resistance of parents and lines to stripe rust at adult stage. (**A**): AvS/AlpF$_7$-71 (*Yr39*); (**B**): Chuanmai 42 (CM42); (**C**): Bainong Aikang 58 (AK58); (**D**): Han 6172 (H6172); (**E**): Zhengmai 9023 (ZM9023); (**F**,**G**): lines from *Yr39*/CM42; (**H**,**I**): lines from *Yr39*/AK58; (**J**,**K**): lines from *Yr39*/H6172; and (**L**,**M**): lines from *Yr39*/ZM9023.

Table 3. Numbers of wheat lines in different stripe rust infection type (IT) categories developed from four backcrosses.

Cross	Resistant	Intermediate	Susceptible
	(IT 1–3)	(IT 4–6)	(IT 7–9)
CM42//CM42/*Yr39*	17	0	0
AK58//AK58/*Yr39*	11	0	0
H6172//H6172/*Yr39*	17	9	0
ZM9023//ZM9023/*Yr39*	15	2	0

3.3. Agronomic Traits

In the crop seasons 2019–2020 and 2020–2021, the mean plant heights of the 5 parents, *Yr39*, CM42, AK58, H6172 and ZM9023, were 99.5, 86.7, 67, 88, and 89.3 cm, respectively (Table 4). Among the progeny lines, the PH values were mainly in the range of 80–100 cm (Figure 2, Table 4). The mean SGN values of the 5 parents were 36, 32, 35, 34, and 34, respectively. Among the 71 progeny lines, 39 had SGN values $\geq$40. Similar results were also obtained for NT among the progeny lines. The mean NT values of *Yr39*, CM42, AK58, H6172 and ZM9023 were 6, 5, 4, 4, and 4, respectively, and the mean NT values of the four backcrosses were between 4 and 6. In terms of TGW, *Yr39* had the lowest mean TGW of 37.7 g, and the mean TGW values of the Chinese cultivar parents were 42.8 g (CM42), 45.5 g (AK58), 46.2 g (H6172), and 42.4 g (ZM9023). The mean TGW values of the inbred lines from the four backcrosses were mainly between 35 g and 50 g.

Table 4. Evaluation of resistance to stripe rust, agronomic traits, and molecular marker detection of the selected wheat lines that the values are the average of the 2019–2020 and 2020–2021 seasons [a].

Parent/Line	Cross	Stripe Rust		Agronomic Trait					Marker		
		IT	DS (%)	PH (cm)	NT	SL (cm)	SGN	TGW (g)	*Xwgp36*	*Xwgp43*	*Xwgp44*
Yr39		2	10	99.5	6	8.3	36	37.7	+	+	+
CM42		7	60	86.7	5	9.8	32	42.8	–	–	–
AK58		8	70	67	4	8.9	35	45.5	–	–	–
H6172		6	70	88	4	9.5	34	46.2	–	–	–
ZM9023		7	60	89.3	4	9.3	34	42.4	–	–	–
SWUST-3 *	CM42/*Yr39*	2	5	88.4	5	9.8	41	42.8	+	+	+
SWUST-4 *	CM42/*Yr39*	2	5	88.6	5	9.2	36	42.2	+	+	+
SWUST-5 *	CM42/*Yr39*	2	5	89.6	7	8.5	51	43.1	+	+	+
SWUST-12	CM42/*Yr39*	2	5	78.8	5	7.9	42	41.5	–	–	+
SWUST-14 *	CM42/*Yr39*	2	20	88.2	5	10.2	39	46.3	+	+	+
SWUST-15	CM42/*Yr39*	2	10	88.4	5	10	40	44.6	–	+	–
SWUST-18	AK58/*Yr39*	2	20	81.8	5	7.9	39	40.7	–	+	+
SWUST-20 *	AK58/*Yr39*	1	5	54.2	5	8.2	35	42.5	+	+	+
SWUST-22 *	AK58/*Yr39*	2	5	68.4	6	8.2	37	47.1	+	+	+
SWUST-24 *	AK58/*Yr39*	1	5	68	7	8.8	37	43.5	+	+	+
SWUST-25 *	AK58/*Yr39*	2	5	65	4	9.9	39	42.2	+	+	+
SWUST-29	H6172/*Yr39*	2	5	80.8	4	11.1	42	43.7	+	–	+
SWUST-33	H6172/*Yr39*	4	30	75.2	5	10.1	44	41.5	+	+	–
SWUST-34	H6172/*Yr39*	3	5	88.4	4	11.2	53	41.3	–	–	+
SWUST-35	H6172/*Yr39*	4	10	90	5	8.4	38	46.7	+	+	–
SWUST-36	H6172/*Yr39*	3	10	89.5	7	9.8	38	43.4	–	–	+
SWUST-44 *	H6172/*Yr39*	2	5	87.8	5	9	35	45.4	+	+	+
SWUST-47 *	H6172/*Yr39*	4	20	88.6	4	8.4	44	43.9	+	+	+
SWUST-52	H6172/*Yr39*	2	30	83.6	5	10.4	48	44.2	+	–	+
SWUST-53	H6172/*Yr39*	3	20	81.8	4	9.7	44	41.8	+	–	+
SWUST-55	ZM9023/*Yr39*	4	30	83.4	5	6.9	35	41.3	+	–	–
SWUST-56	ZM9023/*Yr39*	3	20	25.7	4	7.5	41	45.8	+	–	–
SWUST-58	ZM9023/*Yr39*	2	5	88.4	5	9.9	45	41.6	–	+	–
SWUST-62	ZM9023/*Yr39*	3	20	82.6	5	9.8	41	49.7	–	+	+
SWUST-63	ZM9023/*Yr39*	2	10	82.6	5	10.9	59	50.4	+	+	–
SWUST-64	ZM9023/*Yr39*	3	20	87	5	10.1	48	52.7	–	+	–
SWUST-65	ZM9023/*Yr39*	2	20	82	6	10.1	48	52.4	–	+	–
SWUST-67 *	ZM9023/*Yr39*	3	10	76	5	10.3	47	42.2	+	+	+
SWUST-68	ZM9023/*Yr39*	3	20	59	6	9	41	40	+	–	+

[a] IT: Infection type; DS: disease severity; PH: Plant height; NT: number of tillers; SL: spike length; SGN: spike grain numbers; TGW: thousand grain weight; *, finally selected 11 lines with excellent agronomic traits and high-level resistance to stripe rust.

In order to select lines with good agronomic traits, we used the following criteria: PH $\leq$ 90 cm, NT $\geq$ 4, SGN $\geq$ 35, and TGW $\geq$ 40 g. 6 lines from CM42//CM42/*Yr39* were SWUST-3, SWUST-4, SWUST-5, SWUST-12, SWUST-14, and SWUST-15. 5 lines from AK58//AK58/*Yr39* were SWUST-18, SWUST-20, SWUST-22, SWUST-24, and SWUST-25. 9 lines from H6172//H6172/*Yr39* were SWUST-29, SWUST-33, SWUST-34, SWUST-35, SWUST-36, SWUST-44, SWUST-47, SWUST-52, and SWUST-53. 9 lines obtained from ZM9023//ZM9023/*Yr39* were SWUST-55, SWUST-56, SWUST-58, SWUST-62, SWUST-63, SWUST-64, SWUST-65, SWUST-67, and SWUST-68. Based on these criteria, 29 lines with desirable agronomic traits were selected (Table 4). These selected lines had IT 1–4 and the DS < 30%.

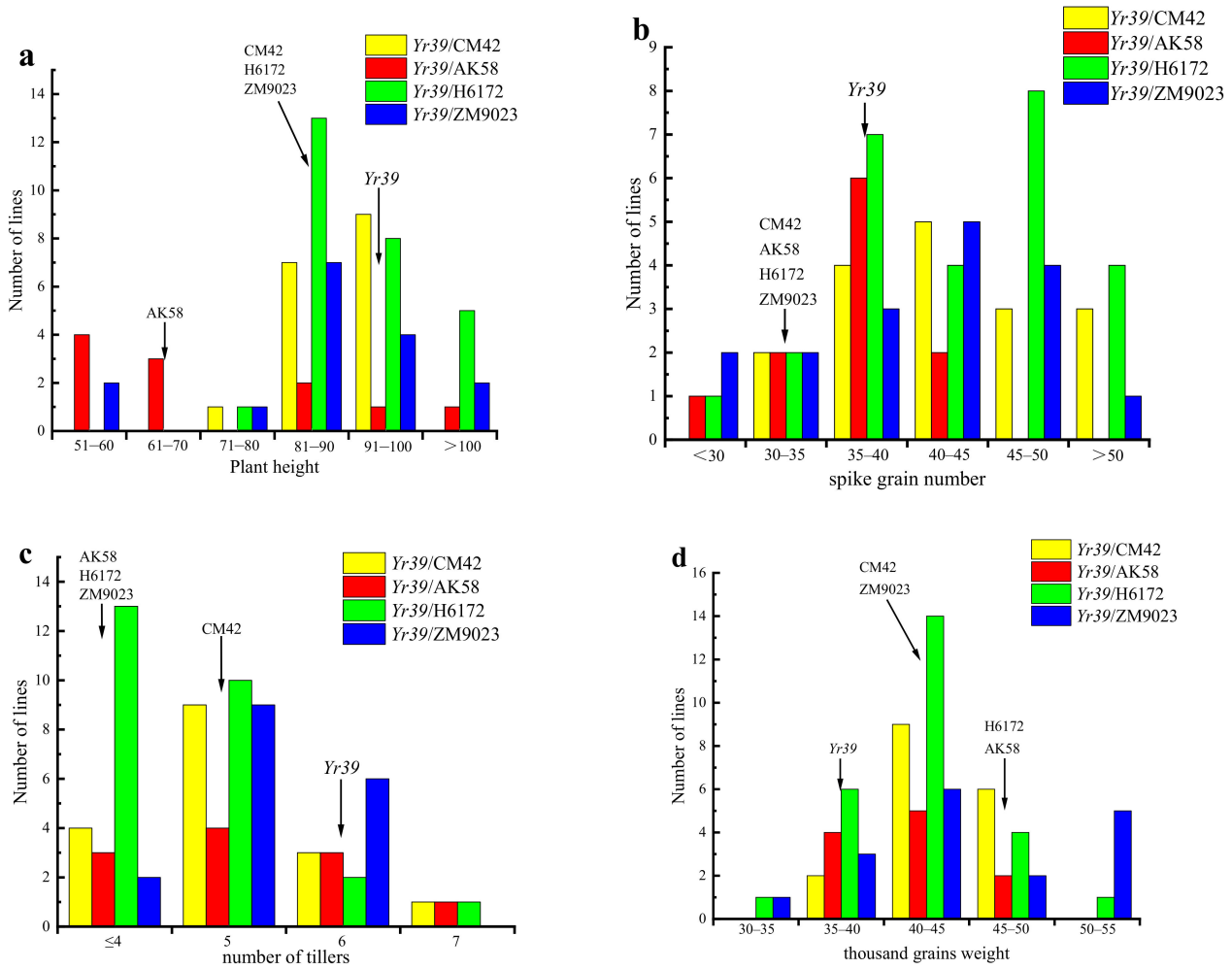

Figure 2. Frequency distribution of plant height (**a**), spike grain number (**b**), number of tillers (**c**) and thousand grains weight (**d**) for 71 backcross progeny lines in four hybrid combinations.

3.4. Molecular Marker Tests

The 71 pre-selected BC_1F_6 lines were tested with *Xwgp36*, *Xwgp43*, and *Xwgp44* (Figure 3). 20 lines (28.16%) had all 3 markers, including 8 from CM42//CM42/*Yr39* (SWUST-3, SWUST-4, SWUST-5, SWUST-6, SWUST-7, SWUST-11, SWUST-14, SWUST-16), 5 from AK58//AK58/*Yr39* (SWUST-20, SWUST-22, SWUST-23, SWUST-24, SWUST-25), 5 from H6172//H6172/*Yr39* (SWUST-30, SWUST-44, SWUST-47, SWUST-49, SWUST-54), and 2 from ZM9023//ZM9023/*Yr39* (SWUST-67 and SWUST-69) (Table 4). There were 26 lines with only 2 markers, accounting for 36.62% of the lines, including 9 with *Xwgp36* and *Xwgp43*, 12 with *Xwgp36* and *Xwgp44*, and 5 with *Xwgp43* and *Xwgp44*. There were 19 lines (26.76%) with only one marker. In addition, 6 lines (8.45%) did not have any of the markers but were resistant to stripe rust, including 2 from CM42//CM42/*Yr39* (SWUST-2 and SWUST-10); one from AK58//AK58/*Yr39* (SWUST-26), one from H6172//H6172/*Yr39* (SWUST-37), and 2 from ZM9023//ZM9023/*Yr39* (SWUST-66 and SWUST-70).

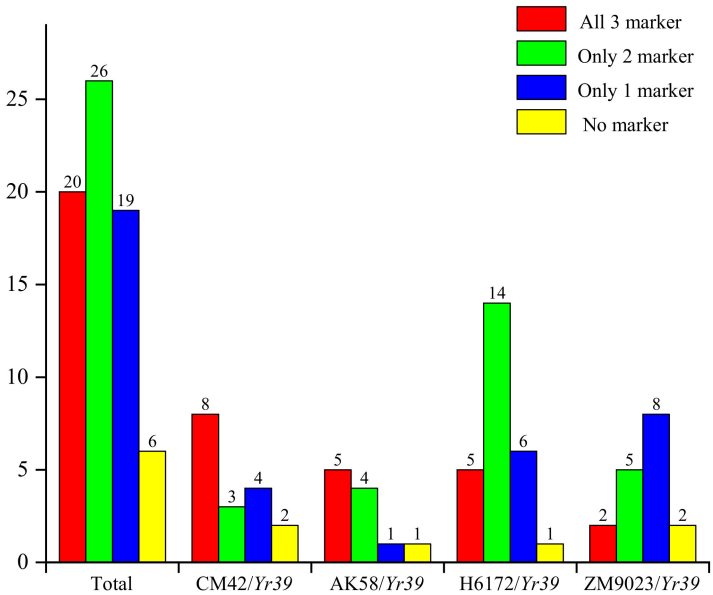

Figure 3. Statistics of marker detection results of 71 lines in 4 backcross progeny lines.

In summary, 20 progeny lines were identified to have markers *Xwgp44*, *Xwgp36*, and *Xwgp43*. Using PH $\leq$ 90 cm, NT $\geq$ 4, SGN $\geq$ 35, and TGW $\geq$ 40 g as the standard for agronomic traits, 29 lines were identified. When the marker data were combined with the agronomic traits, 11 lines with desirable agronomic traits and high-level resistance to stripe rust were finally selected. The characterization of the agronomic traits as well as stripe rust resistance based on both marker genotypes and phenotypes are presented in Table 4.

4. Discussion

4.1. Effectiveness of Yr39 HTAP Resistance in China

Yr39 is a HTAP resistance gene identified from the spring wheat cultivar Alpowa developed by Washington State University and grown in the US Pacific Northwest [19]. Based on Lin and Chen (2007), Alpowa was susceptible at the seedling stage and at the adult-plant stage at the low-temperature profile (4–20 °C) but was resistant (IT 2–5) at the adult-plant stage at the high-temperature profile (10–30 °C) [19]. Our results confirmed that Alpowa has HTAP resistance to stripe rust in fields in Sichuan and with Chinese *Pst* races. We have evaluated the resistance of *Yr39* in the field for the past eight years, and the results showed that the *Yr39* line was consistently resistant during the adult-plant stage in the greenhouse tests with predominant races CYR32, CYR33, and CYR34 and in the fields inoculated with these races or under natural infection (data not shown). Thus, the *Yr39* gene confers durable non-race specific resistance and can be used to breed stripe rust resistant cultivars.

4.2. Evaluation of Molecular Markers Linked with Yr39

In the marker tests of the progeny lines, the percentages of *Xwgp36*, *Xwgp44*, and *Xwgp43* markers in the selected 71 transfer lines were 67.6, 60.6, and 56.3%, respectively. The percentage of only two of the markers was 36.62%, among which the percentage of *Xwgp36* and *Xwgp44* was 16.90%, *Xwgp36* and *Xwgp43* was 12.68%, and *Xwgp43* and *Xwgp44* was 7.04%. The percentage of all three markers in the progeny lines was 28.17%. Some lines had only one marker but showed strong resistance. Peng et al. (2000) reported that when the target gene and the adjacent marker were less than 2 cm, the identification efficiency could reach more than 90% [29]. The genetic distances of *Xwgp36*, *Xwgp44*, and *Xwgp43* to *Yr39* were estimated at about 0.5 cm, 2.7 cm, and 6.6 cm, respectively [19]. Only the genetic distance of *Xwgp36* to *Yr39* was less than 2 cm, and the progeny lines with only *Xwgp36* showed good resistance to stripe rust. *Xwgp44* and *Xwgp43* were more than 2 cm away from *Yr39*, but the progeny lines of only *Xwgp44* or *Xwgp43* detected also showed

good resistance, similar to those with only *Xwgp36*. This phenomenon was found in all four backcrosses. As the present study could not determine which marker was tightly linked to *Yr39*, we used a more conservative detection range to ensure the successful transfer of *Yr39*. The progeny lines with all 3 tested markers, *Xwgp36*, *Xwgp44*, and *Xwgp43*, were considered as successful transfers of *Yr39*. Based on the field tests of the recipient parents and backcross lines, we speculate that *Yr39* could interact with other unknown resistance genes of the recipient parents for enhancing resistance, and the selected progeny lines would show higher stripe rust resistance than the parents.

In the present study, 6 lines did not have any markers linked with *Yr39*, namely SWUST-2 (IT 2, DS 5) and SWUST-10 (IT 3, DS 10) from CM42//CM42/*Yr39*, SWUST-26 (IT 3, DS 30) from AK58//AK58/*Yr39*, SWUST-37 (IT 4, DS 20) from H6172//H6172/*Yr39*, SWUST-66 (IT 2, DS 5) and SWUST-70 (IT 3, DS 20) from ZM9023//ZM9023/*Yr39*. However, in the field test, only SWUST-37 showed moderate resistance. The other lines showed high resistance and this situation occurred in all 4 crosses. The reasons for this situation could be the following. (1) Gene recombination occurred in the hybridization process, during which *Yr39* was transferred into the recipient genome, but none of the molecular markers were introduced. (2) The markers *Xwgp43* (about 6.6 cm) and *Xwgp44* (approximately 2.7 cm) linked to the *Yr39* gene were too far apart, which increased the probability of recombination between the *Yr39* and the molecular marker loci. (3) The resistance genes of the recipient parents and *Yr39* had an additive effect, providing higher levels of resistance to stripe rust. CM42 was reported to have *Yr26* [22], and ZM9023 was reported to have APR genes *Yr29* and *Yr30* [23]; the other 2 recipient parents were resistant to stripe rust during their development at the time of release, but their resistance genes were not determined. It is necessary to identify the type of resistance and its genes in the recipient parents, so that the additive interactions with *Yr39* and other genes for enhancing stripe rust resistance can be more efficiently used in developing new cultivars.

As one of the major regions with frequent stripe rust epidemics, Sichuan has a unique environment for stripe rust research. Wheat germplasm can be reliably evaluated in the field under the natural infection of *Pst*. Through field phenotyping and molecular marker genotyping, the reliability of the selected lines for resistance to stripe rust can be further improved. At the same time, resistant lines without molecular markers for the target gene but showing resistance can also be identified. In the backcross lines, the higher levels of resistance than conferred by *Yr39* alone could be due to the additive effect of other stripe rust resistance genes in certain recipient lines. It is necessary to map and identify the resistance gene in the recipient parents to confirm this hypothesis. In the present study, RGAP markers, *Xwgp36*, *Xwgp44* and *Xwgp43* developed by Lin and Chen [19] were used to detect *Yr39*, but the detection rate in the transfer population was not high, the repeatability was quite low, and polyacrylamide gel electrophoresis was needed for separating PCR products. It is necessary to develop high-throughput and easier-to-use markers for *Yr39*, such as KASP markers.

4.3. The Effect of Yr39 Transfer on Yield Components

The utilization of disease-resistant resources is often accompanied by yield penalty [30]. Therefore, the yield penalty caused by using disease-resistant resources is one of the main concerns of breeders [31]. Tian et al. proposed that disease is the main driving force of biological evolution, and each *R* gene has its own characteristics, which also leads to the existence of disease-resistance costs and the nature and size of disease-resistance costs [32]. The presence of the cost of resistance may lead to loss of function at other polymorphic loci in other ways. Thousand-grain weight is one of the three components of yield. In the present study, after two years of field trials, the selected 20 progeny lines containing *Yr39*, with the mean TGW values from 38.9 to 47.1 g, were better than the *Yr39* donor parent (37.7 g). More than half of the 20 progeny lines had mean TGW values within the range of the recipient parents (42.4–46.2 g), and 3 lines had TGW superior to the recipient parent (SWUST-11: 46.5 g, SWUST-14: 46.3 g and SWUST-22: 47.1 g). *Yr39* confers durable

resistance to stripe rust, and it has demonstrated stable resistance in lines developed using different Chinese wheat cultivars. The gene did not show significant adverse impact on the TGW (Figure 4). Further tests are needed to determine the grain yield of the selected lines in various wheat-growing areas under various environmental conditions including with or without different levels of stripe rust. Nevertheless, the present study showed that the line *Yr39* can be used as a useful stripe rust resistance resource for developing wheat cultivars with durable resistance.

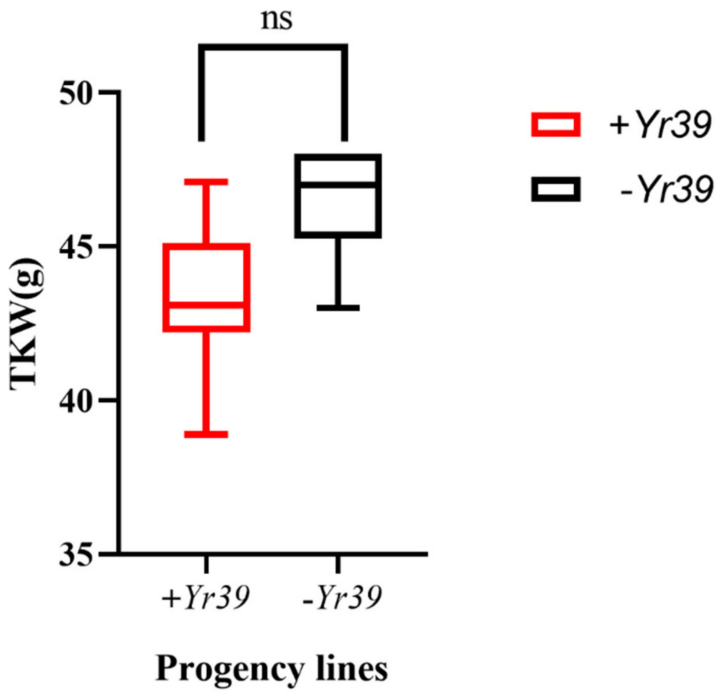

Figure 4. Thousand-seed weight distribution in 4 backcross progeny lines. "+*Yr39*" represents the presence of *Yr39*; "-*Yr39*" represents the absence of *Yr39*; *p*-value was generated by the *t*-test; "ns" represents $p > 0.05$, no significant differences.

5. Conclusions

The 11 wheat lines selected in this study showed high resistance to stripe rust, and excellent agronomic traits. Therefore, they can be used for developing new wheat cultivars with durable stripe rust resistance and high yield potential.

Author Contributions: X.Z. (Xiaochen Zhen) and J.Z. tested markers, analyzed data and wrote the first draft of the manuscript. X.Z. (Xiaochen Zhen), M.Z., W.T., C.M., R.T. and Q.Y. conducted field testing and collected phenotypic data. X.Z. (Xinli Zhou), S.Y., X.L. and C.X. evaluated the population and parents. X.C. and X.Z. (Xinli Zhou) contributed suggestions for the project and revised the first draft. X.C., Z.K. and X.Z. (Xinli Zhou) conceived the project and produced the final version of the manuscript. All authors have read and agreed to the published version of the manuscript.

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Conflicts of Interest: The authors declare no conflict of interest.

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